Year 7 Mathematics

Unit 2: Number Skills

2.2 Factors, Multiples and Primes

Factors, Multiples and Primes

STARTER · 5 MIN

Date: _____ **Title: Factors, Multiples and Primes**

LEARNING OBJECTIVES

- To understand the difference between factors, multiples and primes

STARTER *Answer in your book. No calculators.*

1. Write down the factors of 15
2. Write the first five multiples of 7
3. Is 23 a prime number?
4. Is 42 a multiple of 7?

Starter — Answers

STARTER

Tick or correct in green pen

1. **1, 3, 5, 15**
2. **7, 14, 21, 28, 35**
3. **Yes: factors 1 and 23**
4. **Yes: 6 × 7 = 42**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the factors of a number	☐
SILVER	I can find the multiples of a number and identify prime numbers	☐
GOLD	I can explain the difference between factors, multiples and primes	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Factors, Multiples and Primes

INTRODUCTION · 3 MIN

Factors divide into a number. Multiples are in its times table.

Factor

Divides exactly into a number.

Factors of 12: 1, 2, 3, 4, 6, 12

Multiple

The number times 1, 2, 3, ... (never ends).

Multiples of 6: 6, 12, 18, 24, ...

Prime number

Exactly two factors: 1 and itself.

5 → 1 and 5. 1 is not prime.

Prime factor

A factor that is also prime.

Prime factors of 12: 2 and 3

Examples 1 and 2 — Finding Factors

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Factors come in pairs 1 and the number are factors

Example 1

Write down all the factors of 12.

Example 2

Write down all the factors of 30.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Find the factor pairs

$1 \times 12,\ \ 2 \times 6,\ \ 3 \times 4$

Factors = 1, 2, 3, 4, 6, 12

Tip: stop when the pairs meet.

Example 2

Find the factor pairs

$1 \times 30,\ \ 2 \times 15,\ \ 3 \times 10,\ \ 5 \times 6$

Factors = 1, 2, 3, 5, 6, 10, 15, 30

Tip: use the divisibility rules to test.

Task 1

PRACTICE · 9 MIN

BRONZE *Write down all the factors*

Hint Factors come in pairs 1 and the number are factors

1. Factors of 8
2. Factors of 20
3. Factors of 24
4. Factors of 17
5. Factors of 45

★ Which number under 20 has the most factors?

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. 1, 2, 4, 8

2. 1, 2, 4, 5, 10, 20

3. 1, 2, 3, 4, 6, 8, 12, 24

4. 1, 17

5. 1, 3, 5, 9, 15, 45

★ 12 and 18 (6 factors each)

Examples 3 and 4 — Multiples and Primes

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Multiples: the times table Prime: exactly two factors

Example 3

Write the first five multiples of 6.

Is 48 a multiple of 6?

Example 4

Which of these are prime?

9 11 15 17 21

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Multiples: the 6 times table

Multiples = 6, 12, 18, 24, 30

48 = 6 × 8, so yes.

Tip: multiples go on for ever.

Example 4

Prime: exactly two factors

$9 = 3 \times 3$, $15 = 3 \times 5$, $21 = 3 \times 7$

Primes = 11 and 17

Tip: 1 is not prime: it has one factor.

Task 2

PRACTICE · 9 MIN

SILVER *Multiples and primes*

Hint Multiples: the times table Prime: exactly two factors

1. First five multiples of 8

2. Is 54 a multiple of 9?

3. Is 1 a prime number?

4. The primes between 20 and 30

5. Which are prime?

   2 9 13 21 27

★ Is 91 prime?

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 8, 16, 24, 32, 40

2. Yes: 9 × 6 = 54

3. No: it has one factor

4. 23 and 29

5. 2 and 13

★ No: 91 = 7 × 13

Examples 5 and 6 — HCF and LCM

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

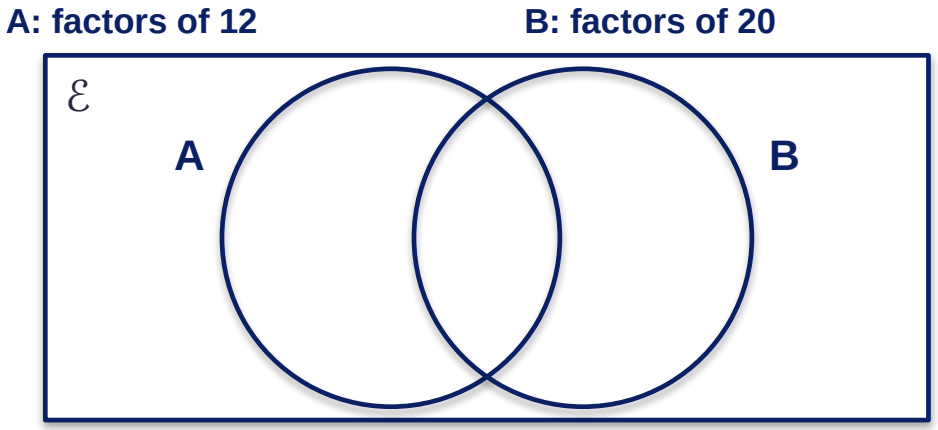
Hint
HCF: highest common factor
LCM: lowest common multiple

Example 5

(3 marks)

Put the factors of 12 and 20 in the Venn diagram.

Write down the HCF.



Example 6

(2 marks)

Bus A: every 10 minutes.
Bus B: every 15 minutes.
Both leave at 08:00.

When do they next leave together?

Examples 5 and 6 — Corrections

CORRECTIONS

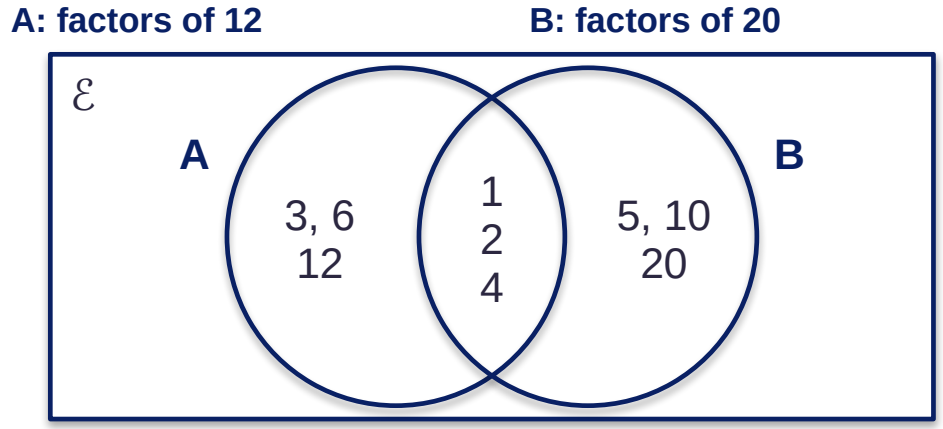
GOLD

Check your copy and correct in green pen

Example 5

Common factors go in the middle

HCF = 4



Tip: HCF = highest common factor.

Example 6

Find the lowest common multiple

10: 10, 20, 30, 40

15: 15, 30, 45

LCM = 30

Time = 08:30

Tip: LCM = lowest common multiple.

Task 3

PRACTICE · 9 MIN

GOLD *Find the HCF or LCM*

Hint HCF: highest common factor LCM: lowest common multiple

1. HCF of 8 and 12

2. HCF of 24 and 36

3. LCM of 4 and 6

4. LCM of 6 and 10

5. Lights flash every 6 s and 8 s. When do they flash together?

★ Explain the difference between a factor and a multiple.

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **HCF = 4**
2. **HCF = 12**
3. **LCM = 12**
4. **LCM = 30**
5. **LCM = 24 seconds**

★ **A factor divides into it;**
a multiple is in its times table.

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write down all the factors of 16.

SILVER Is 29 a prime number? Explain.

GOLD Find the HCF of 18 and 27.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **1, 2, 4, 8, 16**

SILVER **Yes: its only factors are 1 and 29**

GOLD **HCF = 9**

Exam Practice — Factors

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint
Factors come in pairs
1 and the number are factors

Question 1

(2 marks)

Write down all the factors of 28.

.....

Question 2

(2 marks)

Here is a list of numbers.

6 9 14 21 30

(a) Which are multiples of 3? (1)

.....

(b) Which are factors of 42? (1)

.....

Exam Practice — Multiples and Primes

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint
Multiples: the times table
Prime: exactly two factors

Question 3

(2 marks)

Circle the prime numbers.

7 12 19 25 31

.....

Question 4

(2 marks)

Multiples of 7.

(a) Write the first five multiples of 7. (1)

.....

(b) Is 56 a multiple of 7? Explain. (1)

.....

Exam Practice — HCF and LCM

EXAM PRACTICE · 10 MIN

GOLD

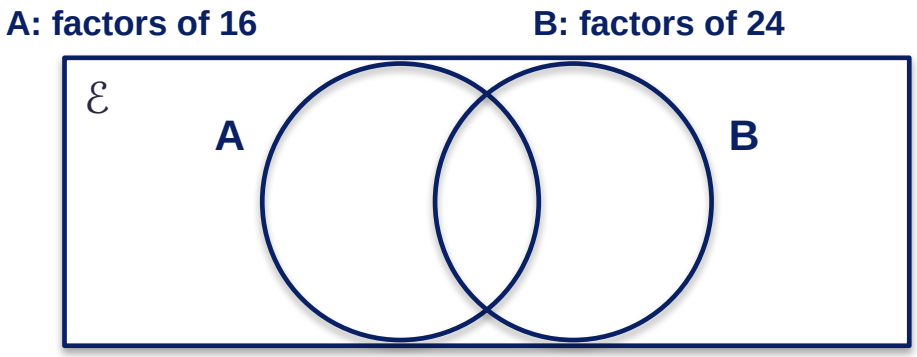
IGCSE exam-style: show all your working

Hint
HCF: highest common factor
LCM: lowest common multiple

Question 5

(3 marks)

Factors of 16 and 24.



(a) Complete the Venn diagram. (2)

(b) Write down the HCF. (1)

.....

Question 6

(2 marks)

Two bells ring together at 10:00.
One rings every 8 minutes and the other every 12 minutes.

When will they next ring together?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

1, 2, 4, 7, 14, 28 B2

B1 for 4 or 5 correct

Tip: pairs 1 × 28, 2 × 14, 4 × 7.

Question 2

(a) 6, 9, 21, 30 B1

(b) 6, 14, 21 B1

Tip: 42 = 6 × 7 = 2 × 21 = 3 × 14.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

7, 19, 31 B2

B1 for two correct, none wrong

Tip: 12 and 25 have more than two factors.

Question 4

(a) 7, 14, 21, 28, 35 B1

(b) Yes: 7 × 8 = 56 B1

Tip: multiples are in the times table.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) A only: 16 — B2

Middle: 1, 2, 4, 8

B only: 3, 6, 12, 24

(b) 8 — B1

Tip: write each factor once.

Question 6

8: 8, 16, 24 12: 12, 24 — M1

LCM = 24

Time = 10:24 — A1

Tip: add the LCM to the start time.